Parth Pandya

| 908-502-6740 | pandyanparth@gmail.com | linkedin.com/in/parthpandya5 | github.com/parthpandya5

Education

Rutgers University - The State University of New Jersey New Brunswick, NJ
Bachelor's of Science in Computer Science, Bachelor's of Science in Data Science (CS Track) *September 2023 – May 2027*

- GPA: 3.7 (Dean's List)
- Coursework: Algorithms, Computer Architecture, Regression Methods, Intro to Data Science, Data Management for Data Science, SQL, Data Structures, Discrete Structures, Statistical Inference for Data Science, Data 101

Experience

Software Engineering Intern June 2026 – August 2026
Conduent *Florham Park, NJ*

- Presented an AI modernization POC to CIO & CTO, securing approval to develop it into an active ADS workflow internally.
- Co-developed Microsoft Copilot and internal AI prompts to convert legacy MCP3 code into technical design documentation and guide C# modernization with mapped business logic.
- Evaluated AI-generated design documents and logic maps against 10+ quality criteria; helped resolve 11 discrepancies and achieve 100% schema compliance across validated outputs.

Software Engineering Intern January 2026 – March 2026
SAMA Wellness *Remote*

- Fine-tuned pretrained BERT on GoEmotions (28 emotions), achieving 82% accuracy on GoEmotions test set for a chatbot.
- Built FastAPI REST APIs with Supabase PostgreSQL for chat history and session persistence; observed ∼200 ms request latency.
- Integrated conditional RAG/Gemini fallback and a weighted 15-question intake flow to personalize recommendations without unnecessary LLM calls.

Software Engineering Intern June 2025 – August 2025
Parabolic YC W25 *San Francisco, CA*

- Configured Logstash pipelines to centralize microservice logs in Elasticsearch, enabling searchable debugging across services.
- Built Kibana dashboards that surfaced microservice events and errors to support engineering troubleshooting.
- Collaborated with senior engineers to integrate ELK observability into microservice monitoring workflows.

Projects

Forge Platform | *Go, PostgreSQL, Docker, Kubernetes, Next.js*

- Built a Go deployment control plane to clone GitHub repositories, build Docker images, and orchestrate Kubernetes workloads.
- Designed transactional PostgreSQL job scheduling with worker leases, heartbeats, and retries to support failure recovery in asynchronous deployments.
- Load-tested the PostgreSQL-backed API at 6.48K requests/sec (p95 6 ms, one-project local fixture) and validated artifact rollback in 8.4 s in a local E2E run.

PandasBlocks | *Next.js, TypeScript, Python, FastAPI, React Flow*

- Built a Next.js/React Flow ML workflow builder with FastAPI execution, persistence, and 35+ data/ML blocks.
- Built a DAG engine with topological sorting, cycle detection, dependency resolution, and per-node error reporting.
- Integrated scikit-learn training and SQLAlchemy persistence; validated execution through automated tests and Docker Compose smoke checks.

Helix Runtime | *C++20, CMake, CUDA (experimental)*

- Built a CPU-first C++20 inference runtime with causal attention, paged KV caching, and continuous batching; CUDA backend compile-validated only.
- Optimized bulk prompt prefill, reducing median CPU trace time 15.8–51.3% across three documented synthetic workloads.
- Eliminated paged-KV history copies, reducing isolated layer CPU time 13.3–27.7% and 2 KiB–1 MiB of temporary data; validated 199 CPU tests.

Technical Skills

Languages: Python, Java, C, C++, Go, C#, TypeScript, JavaScript, SQL, HTML/CSS
Frameworks & Libraries: Spring Boot, React/React Native, Next.js, Node.js, FastAPI, Streamlit, Kafka, PySpark, Pandas, NumPy, Scikit-learn, Seaborn, TensorFlow, PyTorch, LangChain
Databases & Cloud: PostgreSQL, MySQL, Elasticsearch, Supabase, AWS (EC2, Bedrock)
Tools & Platforms: Git/GitHub, Docker, Kubernetes, Azure DevOps, GitHub Actions, Logstash, Kibana, REST APIs, CI/CD, Unit Testing